Payload Field Map

Version	1.4
Date	23 September 2026
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Status	Generated from delivered payloads; v1.1 added `findings`, `pages`, `verification_status`; v1.2 added `confidence`; v1.3 adds the engine's annual figure and calculation, `rate_unit`, `payment_history`, relationship in Cartograph's vocabulary and `member_status`; v1.4 adds the Work Number header to `vois[]` (`overtime_rate`, `overtime_frequency`, `employment_start_date`), drops the empty VOI row, and gives the questionnaire disclosure findings a category and subject

Every field the audit engine sends today, what each one means, and what it needs on the Cartograph side.

This replaces the ordering in *Schema Change Request v1.1*. Since v1.1 of this document the payload carries `findings` (section 8) and every income and asset record names the packet pages it was read from. What follows is what is arriving now.

Both delivered payloads are already in `raw_payload` and can be read alongside this document. This exists because the payload shows the shape and not the meaning, and several of the meanings decide compliance answers.

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1. Six things that are not visible in the JSON

Read these before the tables. Each is a distinction the payload depends on and does not announce.

`current_value` **and** `manual_balance` **are not two amount columns.** `current_value` is a balance backed by a bank statement or a verification of assets. `manual_balance` is the resident's own figure with no third-party evidence. If they merge, a self-declaration satisfies a checklist item that exists to require verification. On both cases delivered so far, every asset arrived as `manual_balance`, none had supporting evidence.

`null` **is not** `false`**.** `is_disabled` and `full_time_student` are true, false, or null, and null means the packet never stated it. A default that turns null into false converts "unknown" into "not disabled", which is also one of the audit's own findings.

`ssn_last4` **is four digits by design.** Not truncated data. The engine holds full SSNs internally and masks them at every outbound point. Full digits are not sent because `CertHouseholdMember#ssn` has no `encrypts` declaration; that remains open in the schema request.

`ref` **and** `member_ref` **are scoped to one request.** They exist so income and asset records can point at a member inside a single payload. They carry no meaning outside it and should not be persisted as identifiers.

`income_type` **or** `asset_type` **of** `other` **is a vocabulary gap, not a description.** It means the extracted term matched nothing in the supplied picklist. The original value is named in the warnings, and those warnings are the list of terms still to agree.

Absent is not empty. Sections not listed here are omitted rather than sent empty, because an empty array asserts that nothing exists while absence asserts nothing at all. `expense_records` and `findings` are absent for that reason.

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2. Envelope

Sent once at the top of every payload. These identify the case and the run that produced it.

Field	Example	Meaning
`schema_version`	`1.3`	Payload contract version. Changes when the shape changes
`extraction_id`	`ext_0ca98dfd...`	Unique per run. A re-audit of the same case produces a new one, so it identifies the attempt rather than the case
`engine_version`	`idp-1.0.0`	Which build produced the extraction. Worth storing: it is what makes a result reproducible when a mapping question comes up months later
`extracted_at`	`2026-09-07T17:06:12Z`	UTC, ISO 8601. When extraction finished, not when the case was received
`case_ref`	`J-PORT-05754`	**Your** `Job#ref_number`**, echoed back.** This is the field to resolve against
`target.job_id`	`5754`	Echoed from the notification. Null if it was not sent
`target.community_id`	`67`	Echoed from the notification
`target.unit_number`	`3-207`	Echoed from the notification, and separately extracted onto `cert_review` from the certification itself. The two can disagree, which is worth surfacing rather than reconciling silently

Everything in `target` is echoed from what the notification carried, not read from the documents. `cert_review.unit_number` is the extracted one.

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3. `cert_review`

One per case.

Field	Example	Meaning
`cert_type`	`annual`	One of `initial`, `annual`, `ar_self_cert`, `interim`. Omitted entirely if the type cannot be determined — see the AR-SC note in section 7
`effective_date`	`2026-10-01`	ISO. Certification effective date
`unit_number`	`3-207`	As printed on the certification, which may differ from the unit record
`hh_size`	`1`	Count of members sent, not a figure read off the form
`annual_income`	`42180.00`	Household total as the certification declares it

Field	Example	Meaning
annual_assets	915.00	Summed from the asset records sent, since the form carries no total
head_of_household_name	Dolores Ackerman	Convenience field; the authoritative flag is is_hoh on the member
tenant_rent	1180.00	
contract_rent	1180.00	The unit's full rent before assistance, where the form prints one (50059 field 29, lease, 3560-8); omitted otherwise
gross_rent	1245.00	As the form prints it: contract + UA on HUD forms, tenant + UA on a TIC
gross_rent_basis	contract_plus_allowance	Which definition the form's figures settle: contract_plus_allowance or tenant_plus_allowance; omitted when neither
hud_gross_rent	1255.00	Contract rent + UA (50059 box 31); only when the form's figures settle it
tenant_gross_rent	940.00	Tenant rent + UA, no subsidy (TIC gross rent; 50059 total tenant payment, derived)
rent_field_sources	{tenant_rent: {form_type, source_job_document_id, effective_date, pages}, …}	One entry per rent figure sent; derived_from names the figures a computed one came from
utility_allowance	65.00	
max_program_rent	1245.00	Program rent limit. Null when not extracted, which disables the rent-limit checks

All amounts are plain decimal strings: no currency symbol, no thousands separator, two decimal places.

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4. household_members

Field	Example	Meaning
ref	m01	Request-scoped handle. Do not persist
existing_member_id	4629	The id of the member in the request's existing_records_manifest this row corresponds to, or null for a member Cartograph does not hold. Linked on date of birth plus a recognisable name, on a close name alone when either side has no date, or on a near-identical uncontested name when the dates disagree (reported in warnings: one of the two dates was misread or mistyped). Names are matched against the manifest's first_name/last_name and every entry of its aka_names

Field	Example	Meaning
person_id	1981	The manifest member's person_id, echoed with existing_member_id; null when the manifest has none
aka_names	["Victoria Pole"]	The packet's spelling of the name when it is not one the manifest already has for the person (neither its name nor an aka). A suggestion for staff to accept or dismiss, never a rename. Empty otherwise
first_name	Dolores	
middle_name	R.	Often absent
last_name	Ackerman	
date_of_birth	1987-05-19	ISO. Null when the printed date could not be parsed, and reported in warnings
is_hoh	true	Exactly one member carries true
relationship	Minor Child	In Cartograph's vocabulary: Head of Household, Spouse, Co-Head, Minor Child, Other Adult, Live-in Aide, Foster Child, Unborn. The form's word (Dependent, Son, a 50059 code) is mapped by word and by age at the effective date: a child relationship under eighteen is Minor Child, from eighteen Other Adult. A word with no equivalent is sent as printed with a warning
relationship_as_printed	Dependent	The form's own word, unmapped
member_status	Active	Active, or Unborn for an expected child listed on the certification: no name, date of birth or SSN, counted in household size
is_disabled	null	true / false / **null**. See section 1
full_time_student	false	Same three-state rule
ssn_last4	4417	Four digits. See section 1
email	null	
cell_phone	555-0164	As printed; not normalized
sort_order	1	Order on the certification

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5. income_records

One per income source. Paystubs arrive nested underneath rather than as a flat list, because grouping by member and employer happens engine-side.

Field	Example	Meaning
`member_ref`	`m01`	Null when the earner could not be matched to a member, with a warning. The record is still sent, the income is real even when its owner is uncertain
`existing_record_id`	`4329`	The `id` of the manifest income record this row corresponds to: the same member (by `existing_member_id`), the same `income_type` with Cartograph's own aliases applied (`non_federal_wages` and `wages_and_salaries` are one type), and, when the member holds several rows of that type, the closest `source_name`. Null for a record with no counterpart, which is created as before. A member sent without `existing_member_id` never carries record ids
`income_type`	`wages_and_salaries`	From the supplied picklist. `other` means no match; see section 1
`source_name`	`Redwood Facilities Group LLC`	Employer or payer as extracted
`frequency_of_pay`	`bi-weekly`	How often the person is paid. As the record states it; when it does not, the rate's own unit (a monthly benefit is paid monthly). Null for an hourly rate with no stated pay period
`rate_unit`	`hourly`	What `rate_of_pay` is per: `hourly`, `daily`, `weekly`, `bi-weekly`, `semi-monthly`, `monthly`, `quarterly`, `annually`, `per_period`. Null when the record has no rate
`annual_income`	`29133.00`	**The engine's annual figure for this source.** Null when no method could produce one (a record with no amount, a declaration with no period)

Field	Example	Meaning
`calculation`	`{method, annual_income, details, alternatives[]}`	How the figure was reached. `method` is `paystub-based`, `history-based`, `voi-based` or `self-declared`; `details` is the arithmetic in words (`avg(2 stubs) = 560.25 × 52 = 29133.00 — only 2 pay stub(s) in the file`). `alternatives[]` are the other rows the engine computed for the source, each with `status` `audit` (a year-to-date projection run beside the primary), `rejected` (a method that produced an implausible figure, with why) or `historical` (income from a period before the certification). The methods-disagree findings are made from this comparison. The calculation is the one for this record's own income type: a payer that pays a member two incomes (the Social Security Administration paying retirement and SSI) gives two records, each with its own figure
`payment_history`	`[{date, amount}]`	For child support, alimony and benefit payment records: every payment line the document prints, as printed. `date` is null when the line prints no date (a numbered worksheet). The engine annualises from these rows; it never sums them on the document's behalf
`date_received`	`2026-08-04`	ISO. When the verification was received
`employment_start_date`	`null`	ISO
`employment_status`	`Active`	`Active`, `Terminated`, `On Leave`, or null. **Needs a column**
`termination_date`	`null`	ISO. **Needs a column**
`self_declared_amount`	`812.50`	The resident's own figure, from the application, questionnaire or an affidavit, distinct from anything verified. Never a figure read off the certification form
`source_of_declaration`	`Application`	Where the self-declared figure came from
`certification_form`	`HUD-50059`	The certification form the next field was read from: `HUD-50059` or `TIC`. Null when the form is one Cartograph does not take by name (RD 3560-8), and the figure is withheld with it
`certification_amount`	`9750.00`	The annual figure the certification itself prints for this source. Cartograph compares it with its own calculation. A source the form lists and nothing verifies arrives as its own `declared_only` record carrying this figure, so there is a row to attach it to

Field	Example	Meaning
`verification_status`	`verified`	`verified` (read from a third-party document), `declared_only` (the household declared it and nothing in the packet backs it), `verified_not_declared` (a document carries it and the certification does not), `self_certified` (AR-SC)
`pages`	`[20, 21]`	Packet pages this record was read from. Positions in the file Cartograph sent; a reviewer opens the same file and lands on the page
`confidence`	`{score, flag, review[]}`	The engine's confidence in this record: `score` 0 to 1, `flag` green / yellow / red, and `review`, the fields a reviewer should look at with `field`, `flag` and `reason`. See section 8a
`paystubs[]`		`pay_date`, `gross_pay`, `ytd_amount`, `pay_frequency`, `pages`
`vois[]`		`voi_type`, `date_received`, `rate_of_pay`, `rate_unit`, `hours_per_pay_period`, `frequency_of_pay`, `ytd_amount`, `ytd_start_date`, `ytd_end_date`, `overtime_rate`, `overtime_frequency`, `employment_start_date`. Sent only when the verification stated something (a rate, hours, a YTD figure or an overtime rate); a source verified by its stubs alone has no VOI row, so nothing arrives as a placeholder. `frequency_of_pay` here is the pay frequency, read from the stubs first. A Work Number report fills both: its pay period rows are the `paystubs[]`, its header (rate, hours, start date, YTD as of a date, status) is the VOI row
`zero_income`	`null`	Object when a zero-income affidavit was filed

The `annual_income` on each record and the `annual_income` on `cert_review` answer different questions: the record's is what the engine computed from the documents, the review's is what the certification declares. When they differ the `CERT_SUMMARY_INCOME_MISMATCH` finding says by how much.

A source can arrive with paystubs and no verification entry. That is not an error: when an employer verification comes back blank, the manager substitutes paystubs, and the engine reconstructs the source from them. Those records carry no `self_declared_amount` and no verification fields, because there is no third-party document behind them.

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6. `asset_records`

Field	Example	Meaning
member_ref	m01	Null when the owner could not be matched, with a warning
existing_record_id	3519	The manifest asset record this row corresponds to, on the same rule as income: same member, same asset_type, and the closest institution_name when the member holds several of that type. Null when there is none
asset_type	checking	From the supplied picklist. other means no match
institution_name	Meridian Savings Bank	Bank or source
current_value	4830.00	**Verified** balance. Present only with a statement or a VOA behind it. See section 1. A verification request the bank has not answered gives the account no figure at all, and such an account is not sent as a record (the import warnings name it); the audit reports the unanswered request as a finding
manual_balance	915.00	**Self-declared** balance: the resident's own figure from a self-certification or questionnaire. Never a figure read off the certification form
bank_stmt_avg_balance	null	Six-month average where stated, or the mean of the VOA's monthly balances when those were sent instead
annual_income_from_assets	0.00	Only when a document states the annual figure outright. A statement's period interest is sent on the statement row instead, never as this total
interest_type	Percentage	**Needs a column**
interest_rate	0.50	The rate the household itself wrote for the account on its self-certification or application, as a percent (2.50 = 2.5%). Never a rate a bank printed; that travels on the statement or VOA row
percentage_of_ownership	100	Joint ownership share
source_of_declaration	null	
certification_form	HUD-50059	As on income records
certification_cash_value	483.00	The cash value the certification itself prints for this asset
certification_income	0.00	The annual income from the asset the certification prints; a printed 0 is sent as 0.00. annual_income_from_assets stays what a verification or the household's own form states, never the certification's line

Field	Example	Meaning
verification_status	verified	Same values as on income records
pages	[13]	Packet pages this record was read from
confidence	{score, flag, review[]}	As on income records
bank_statements[]		statement_date, balance, and the income the statement shows: interest_earned (interest or dividends for its period, in dollars; never deposits), interest_rate (the rate it prints, as a percent), months_covered (whole months the period spans, 1 for a monthly statement). Only statements carrying a date or a balance are sent; a rate outside 0 to 100 or a loss is withheld with a warning
eivs[]		One row per benefit section of an EIV income report, attached to that benefit's income record: benefit_type (social_security, ssi, dual_entitlement, employment_wages, unemployment_benefits, other), source_name, report_date (the report's own date), amount + frequency (the figure in force), annual_amount (only when the report states one), payment_status as printed, notes, pages. An EIV report is never sent as a vois row
voa	null	voa_date, reported_value, source, interest_income (the annual interest the institution states, in dollars), interest_rate (the rate it states, as a percent), and month_1_balance…month_6_balance (oldest first) when the VOA lists monthly balances instead of an average

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6b. checklist_matches

Field	Example	Notes
finding_id	88412	The case's own checklist row id, from the case request
found	true	The form is in the packet
pages	[2, 3]	Packet pages it was read from
confidence	0.85	Certainty of the match and the note together, 0-1
note	HUD 50059 present, pages 2-3. Not signed: …	For the row's note field; no marker in the text

Field	Example	Notes
`note_source`	`scan`	Always `scan`; what tells an engine note from one staff typed, so Cartograph clears only scan notes on a rerun

Rows the engine cannot map to a document type are not sent.

7. What needs adding

Three columns are being sent right now with nowhere to land:

Table	Column	Type
`cert_income_records`	`employment_status`	string
`cert_income_records`	`termination_date`	date
`cert_asset_records`	`interest_type`	string

Two decisions, neither blocking:

AR-SC — this one does block. `CertReview::CERT_TYPES` allows `initial`, `annual`, `interim`. The engine audits four types, and AR-SC now arrives as `ar_self_cert`, a fourth value. It is not collapsed onto `annual`: AR-SC applies a different rule set — the certification form is the source of truth and no third-party wage verification is expected — and sending it as an ordinary annual produces false findings about documents the file is not supposed to have. `CERT_TYPES` needs the fourth value before an AR-SC case can import.

An unmappable or missing type is now **omitted** rather than defaulted. `cert_type` selects the checklist template through `cert_type_scope`, so a guessed value produces a clean-looking audit against the wrong rule set. A missing one is visible; a plausible wrong one is not.

`source` **on the cert child tables.** Re-import clears extraction-owned records and rewrites them. Without a marker separating what the importer created from what an analyst typed, the delete removes both. This only matters on the second import of a case, by which point reviewer work is in place. Until it exists, treat a cert review that already has records as a conflict rather than proceeding.

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8. `findings`

One object per finding of the audit, in the shape of the Scan Findings thread. Field-level review notes ("Review recommended" on one field of one record) are the scorer's commentary, not findings, and are not sent.

Field	Example	Meaning
`finding_key`	`ASSET_SELF_DECLARED_VS_VERIFIED:randy_buck:savings`	Stable identity: code plus the subject. A re-scan that reports the same finding sends the same key, so it updates the row rather than opening another
`code`	`SIGNATURE_DATE_MISSING`	The finding type. `NOTE` marks a finding that is still plain text; its key is a hash of the wording with digits removed, so a re-read figure updates the same row
`category`	`income`	`file_review`, `income`, `asset`, `household_member`, `unit_rent`

Field	Example	Meaning
`result`	`non_compliant`	`non_compliant` is an issue to review. `compliant` and `na` are informational (a name spelled two ways, an amount that is a subtotal) and should not ask for a verdict
`subject_type`	`income_record`	`household_member`, `income_record`, `asset_record`, `document`, or null for the case
`subject_label`	`Randy Buck — Midstates Bank`	Human-readable subject
`member_ref`	`m01`	The member the finding is about, when it names one
`pages`	`[13]`	Where it was found. Empty when the finding is about the case as a whole (a missing form)
`label`	`Self-declared asset balance differs from the verified balance`	Short title
`description`		The full wording
`correction_required`		What resolves it, when the rule knows
`assignment`	`client`	`internal`, `client`, `procedural_issue`, or null
`resolution_type`	`recalculation`	`presence_only` or `recalculation`
`disputes_extraction`	`true`	The finding says the extraction contradicts the packet; these lower the engine's own confidence

Every scan sends the complete set. A key present in an earlier scan and absent from the latest one is cleared, not still open.

Category and `subject_type` are the key for the next action. A finding whose category is `asset` and whose subject is `asset_record` with no `subject_label` is a record that should exist and does not; its `correction_required` says so ("Add an asset record for the disclosed life insurance …"). The questionnaire disclosure findings arrive this way as of 23 September: `QUESTIONNAIRE_LIFE_INSURANCE_UNVERIFIED`, `_REAL_ESTATE_`, `_CHECKING_` and `_SAVINGS_` as `asset` / `asset_record`; `_EMPLOYMENT_`, `_SSA_`, `_CHILD_SUPPORT_` and `_PENSION_` as `income` / `income_record`; `QUESTIONNAIRE_STUDENT_UNVERIFIED` as `file_review` with no subject, since the missing item is a certification and not a record. Each is assigned to the client and keyed per case, so a re-scan updates the row.

Verdicts come back on a separate `findings_feedback` event, nightly, one per case: `case_ref`, `scan_id`, then `verdicts[]` of `finding_key`, `verdict` (`valid` / `invalid`), `verdict_reason`, and `manual_findings[]` of `description`, `page`, `subject_label`, `source: "manual"`, `matched_checklist_item` (key and name). The import-result callback is not the place for them: it arrives once at import time and a later call on it would overwrite the import outcome.

`expense_records` is still not sent. The engine does not extract expenses.

8a. `confidence`

On every household member, income record and asset record, on `cert_review`, and once at the top level for the case.

Field	Example	Meaning
`score`	`0.79`	0 to 1. Green is 0.80 and above, yellow 0.50 and above, red below. A value never found on a source page is capped at 0.79, so green means "found in the document"
`flag`	`yellow`	What a reviewer triages by. A record the household only declared, or one a finding disputes, cannot be green
`review`	`[{"field": "rateOfPay", "flag": "yellow", "reason": "Found only on the certification form, not in this record's documents"}]`	The fields to look at, each with the reason. Empty on a green record

The top-level object carries `score`, `flag` and the field counts (`green`, `yellow`, `red`, `na`). The case flag is bounded by the worst disputed record and is yellow whenever any income is declared-only; it is a triage signal, not an average.

Green fields can go straight in. Yellow and red are the review list; the reason says what to check.

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9. On the warnings

Every payload carries a list of what the mapping could not carry: a date that would not parse, an earner who matched no member, a type with no equivalent in the picklist. They are the only record of where the two vocabularies disagree, and the accumulated list is the agenda for closing the gaps.

The same applies in the other direction. Please populate `warnings` on the import result for every value that collapses, every date that fails to parse, and every member that arrives without a relationship, without them, that information is recoverable only by manual spot-checking.